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Puncture needles

Abstract

A puncture needle can comprise an elongate portion, a puncture component associated with a distal end of the elongate portion, and a spacer associated with a distal portion of the elongate portion, where the spacer can be configured to contact a wall of the delivery catheter lumen to prevent contact between the puncture component and the wall. A puncture needle can comprise an elongate portion and a puncture component associated with a distal end of the elongate portion, where the puncture component comprises a blade edge on a distal edge, and where the puncture component is configured to be advanceable through a lumen of a delivery catheter without the blade edge sticking to a wall of the lumen.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of PCT International Patent Application Serial No. PCT/US2020/045713, filed Aug. 11, 2020 and entitled PUNCTURE NEEDLES, which claims the benefit of U.S. Provisional Patent Application No. 62/890,528, filed Aug. 22, 2019 and entitled PUNCTURE NEEDLES, the complete disclosures of both of which are hereby expressly incorporated by reference herein in their entireties.

BACKGROUND**Field**

(1) The present disclosure generally relates to the field of transcatheter delivery of medical implant devices and/or therapies.

Description of Related Art

(2) Puncture needles can be used in minimally invasive transcatheter approaches to enable delivery of medical implant devices and/or therapies to a target tissue site. Formation of an opening in the tissue of an internal vessel, channel, organ and/or chamber to allow delivery of medical implant devices and/or therapies thereto can require transcatheter delivery of a puncture needle to the target tissue site.

SUMMARY

(3) Described herein are one or more devices, systems and methods relating to puncture needles used in minimally invasive transcatheter approaches, where the puncture needles can be advanced through a delivery catheter lumen without becoming jammed within the lumen, including through one or more bends in the delivery catheter lumen.

(4) For purposes of summarizing the disclosure, certain aspects, advantages and novel features have been described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, the disclosed embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

- (5) In some implementations, a puncture system can comprise a puncture needle configured to be slidably advanced through a delivery catheter lumen, the puncture needle comprising an elongate portion, and a puncture component associated with a distal end of the elongate portion, the puncture component being configured to puncture tissue. A spacer can be associated with a distal portion of the elongate portion, wherein the spacer can be configured to contact a wall of the delivery catheter lumen to prevent contact between the puncture component and the wall when the puncture component is advanced through a bend in the delivery catheter lumen.
- (6) In some embodiments, the spacer comprises a predetermined lateral dimension and a predetermined longitudinal dimension configured to prevent the contact between the puncture component distal end and the wall.
- (7) In some embodiments, the spacer comprises a distal end, a proximal end, and a lateral surface between the distal end and the proximal end, wherein the lateral surface is at a predetermined lateral distance from the distal portion of the elongate portion, at least one of the lateral surface and the distal end being configured to contact the wall of the delivery catheter lumen to prevent the contact between the puncture component distal end and the wall.
- (8) In some embodiments, the lateral surface comprises a concavely curved surface configured to contact the wall of the delivery catheter lumen. In some embodiments, the lateral surface comprises a convexly curved surface configured to contact the wall of the delivery catheter lumen. In some embodiments, the lateral surface comprises a surface parallel to a longitudinal axis of the puncture needle, the surface being configured to contact the wall of the delivery catheter lumen.
- (9) In some embodiments, a longitudinal cross section of the spacer comprises at least one of a segment of a circle, a segment of an oval, and a segment of a teardrop shape, wherein the longitudinal cross section is along a plane extending laterally from the distal portion of the elongate portion along a longitudinal axis of the puncture needle.
- (10) In some embodiments, the spacer is symmetrical around an axis collinear with a longitudinal axis of the puncture needle. In some embodiments, the spacer is asymmetrical around an axis collinear with a longitudinal axis of the puncture needle.
- (11) In some embodiments, the spacer extends partially circumferentially around the distal portion of the elongate portion. In some embodiments, the spacer is positioned circumferentially around the distal portion of the elongate portion.
- (12) In some embodiments, the spacer comprises a ring shape. In some embodiments, a lateral surface of the ring shape comprises a concave curvature. In some embodiments, a lateral surface of the ring shape comprises a convex curvature. In some embodiments, a lateral surface of the ring shape is parallel to an axis collinear with a longitudinal axis of the puncture needle.
- (13) In some embodiments, the spacer comprises a plurality of distinct portions positioned around the distal portion of the elongate portion. In some embodiments, the spacer comprises a plurality of lobes positioned circumferentially around the distal portion. In some embodiments, the spacer comprises a plurality of panels extending circumferentially around the distal portion of the elongate portion.
- (14) In some embodiments, the spacer is adjacent to the puncture component. In some embodiments, the spacer is a predetermined distance proximal of the puncture component.
- (15) In some implementations, a method of puncturing tissue can comprise inserting a puncture needle into a lumen of a delivery catheter, wherein the puncture needle can comprise an elongate portion, a puncture component associated with a distal end of the elongate portion, and a spacer associated with a distal portion of the elongate portion. The method can comprise advancing the puncture component of the puncture needle through a bend in the lumen of the delivery catheter, contacting a wall of the lumen of the delivery catheter with a portion of the spacer associated with the distal portion of the elongate portion to prevent contact between the puncture component and the wall when the puncture component is advanced through the bend in the delivery catheter, extending the puncture component through an outlet opening of the delivery catheter, and

puncturing tissue at a target site using the puncture component to form an opening in the tissue.

(16) In some embodiments, the method can comprise enlarging the opening in the tissue, wherein enlarging the opening comprises inserting at least a portion of the spacer into the opening.

(17) In some embodiments, puncturing tissue at the target site comprises puncturing tissue at a left atrial wall location from within a coronary sinus. In some embodiments, the method can comprise inserting the delivery catheter transfemorally into a right atrium and from the right atrium into a coronary sinus through a coronary sinus ostium.

(18) In some implementations, a puncture needle can comprise an elongate portion, and a puncture component associated with a distal end of the elongate portion, the puncture component comprising a blade edge on a distal edge, the blade edge being configured to puncture tissue at a target site, wherein the puncture component is configured to be advanced through a lumen of a delivery catheter without the blade edge sticking to a wall of the lumen at a bend in the delivery catheter.

(19) In some embodiments, the blade edge is on a portion of the distal edge, the distal edge comprising a blunt portion.

(20) In some embodiments, the blade edge comprises a convex curvature. In some embodiments, the blade edge is a convexly-curved blade edge. In some embodiments, the blade edge comprises a segment of a circle. In some embodiments, the blade edge comprises a segment of an oval. In some embodiments, the blade edge comprises a linear blade edge between opposing curved corners.

(21) In some embodiments, a portion of the puncture component proximal of the blade edge has a width wider than a width of the blade edge. In some embodiments, a cross section of a distal portion of the puncture component taken along a plane extending between opposing portions of a pair of lateral surfaces comprises a segment of a circle, the blade edge being on a distal portion of the segment of the circle, and wherein opposing portions of the segment of the circle proximal of the blade edge are blunt.

(22) In some embodiments, a proximal portion of the puncture component has a width narrower than a width of the blade edge. In some embodiments, a proximal portion of the puncture component has a width the same as a width of the blade edge.

(23) In some embodiments, a proximal portion of the puncture component has a height the same as that of a distal portion of the puncture component. In some embodiments, a proximal portion of the puncture component has a height larger than that of a distal portion of the puncture component.

(24) In some implementations, a method of puncturing tissue can comprise inserting a puncture needle through a lumen of a delivery catheter. The puncture needle can comprise an elongate portion and a puncture component associated with a distal end of the elongate portion, and the puncture component can comprise a blade edge on a distal edge of the puncture component. The method can comprise advancing the puncture needle through a bend in the delivery catheter without the puncture component sticking to a wall of the lumen of the delivery catheter, extending the puncture component through an outlet opening on the delivery catheter, and puncturing tissue at a target site using the blade edge to form an opening at the target site.

(25) In some embodiments, the method can comprise enlarging the opening in the tissue, wherein enlarging the opening comprises further inserting the puncture component into the opening.

(26) In some embodiments, a cross section of a distal portion of the puncture component taken along a plane extending between opposing portions of a pair of lateral surfaces comprises a segment of a circle, the blade edge being on a distal portion of the segment of the circle, and wherein opposing portions of the segment proximal of the blade edge are blunt portions, and wherein enlarging the opening comprises inserting the blunt portions into the opening to atraumatically enlarge the opening.

(27) In some embodiments, puncturing tissue at the target site comprises puncturing tissue at a left atrial wall location from within a coronary sinus. In some embodiments, the method can comprise inserting the delivery catheter transfemorally into a right atrium and from the right atrium into a coronary sinus through a coronary sinus ostium.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various embodiments are depicted in the accompanying drawings for illustrative purposes and should in no way be interpreted as limiting the scope of the inventions. In addition, various features of different disclosed embodiments can be combined to form additional embodiments, which are part of this disclosure. Throughout the drawings, reference numbers may be reused to indicate correspondence between reference elements. However, it should be understood that the use of similar reference numbers in connection with multiple drawings does not necessarily imply similarity between respective embodiments associated therewith. Furthermore, it should be understood that the features of the respective drawings are not necessarily drawn to scale, and the illustrated sizes thereof are presented for the purpose of illustration of inventive aspects thereof. Generally, certain of the illustrated features may be relatively smaller than as illustrated in some embodiments or configurations.
- (2) FIG. 1 shows examples of transcatheter pathways to access the heart.
- (3) FIG. 2 is a side view of an example of a puncture needle and an associated spacer positioned within a delivery catheter lumen.
- (4) FIGS. 3A and 3B show a perspective view and a longitudinal cross section view of the puncture needle and spacer of FIG. 2, respectively.
- (5) FIGS. 4A and 4B show a perspective view and a longitudinal cross section view, respectively, of an example of a puncture needle and an associated spacer which comprises an outwardly curving lateral surface.
- (6) FIGS. 5A and 5B show a perspective view and a longitudinal cross section view, respectively, of an example of a puncture needle and an associated spacer which comprises an inwardly curving lateral surface.
- (7) FIGS. 6A and 6B show a perspective view and a longitudinal cross section view, respectively, of an example of a puncture needle and an associated spacer comprising an outwardly curving lateral surface, where a distal portion of the lateral surface has a smaller radius of curvature than a proximal portion of the lateral surface.
- (8) FIGS. 7A and 7B show a perspective view and a longitudinal cross section view, respectively, of an example of a puncture needle and an associated spacer comprising an outwardly curving lateral surface, where a proximal portion of the lateral surface has a smaller radius of curvature than a distal portion of the lateral surface.
- (9) FIGS. 8A and 8B show a perspective view and a longitudinal cross section view, respectively, of an example of a puncture needle and an associated spacer which is asymmetrical around an axis collinear with a longitudinal axis of the puncture needle.
- (10) FIGS. 9A and 9B show a perspective view and a longitudinal cross section view, respectively, of an example of a puncture needle and an associated spacer which includes a plurality of panels positioned around an elongate portion of the puncture needle.
- (11) FIGS. 10A, 10B, and 10C show a perspective view, a view from a distal end, and a longitudinal cross section view, respectively, of an example of a puncture needle and an associated spacer which includes a plurality of distinct outwardly curving portions circumferentially positioned around an elongate portion.
- (12) FIG. 11 is a flow diagram of an example of a process for deploying a puncture needle comprising a spacer as described herein.
- (13) FIG. 12 is a longitudinal cross-sectional view of an example of a puncture needle system comprising a protective cover for a distal end of the puncture needle.
- (14) FIG. 13 is a side view of an example of a puncture needle which comprises a puncture component with a curved blade edge positioned within a delivery catheter lumen.

(15) FIGS. 14A, 14B and 14C are a perspective view, a top-down plan view, and a side view, of the puncture needle of FIG. 13, respectively.

(16) FIG. 15 is a perspective view of an example of a puncture needle comprising a puncture component with a curved blade edge.

(17) FIGS. 16, 17, 18 and 19 are top-down plan views of examples of puncture components with a curved blade edge.

(18) FIGS. 20, 21, 22 and 23 are top-down plan views of examples of puncture components with a blade edge which includes a linear or substantially linear portion extending between curved corners.

(19) FIGS. 24 and 25 are longitudinal cross-sectional views of examples of puncture components comprising a blade edge.

(20) FIG. 26 is a flow diagram of an example of a process for deploying a puncture needle which has a puncture component with a blade edge as described herein.

DETAILED DESCRIPTION

(21) The headings provided herein are for convenience only and do not necessarily affect the scope or meaning of the claimed invention.

(22) The present disclosure relates to systems, devices, and methods relating to minimally invasive transcatheter delivery of medical implant devices and/or therapies to target tissue locations.

Described herein are puncture needles which can be advanced from a proximal portion, including a proximal end, to a distal portion, including a puncture needle outlet opening of the distal portion, of a lumen of a delivery catheter positioned within a tortuous anatomical pathway. The puncture needles can be used to pierce tissue at the target tissue locations to enable delivery of the medical implant devices and/or therapies to the target tissue locations. The puncture needles can be advanced through the delivery catheter lumen without becoming jammed within the lumen, including through one or more bends and/or curvatures in the delivery catheter lumen.

(23) Although certain preferred embodiments and examples are disclosed below, inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses and to modifications and equivalents thereof. Thus, the scope of the claims that may arise herefrom is not limited by any of the particular embodiments described below. For example, in any method or process disclosed herein, the acts or operations of the method or process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various operations may be described as multiple discrete operations in turn, in a manner that may be helpful in understanding certain embodiments; however, the order of description should not be construed to imply that these operations are order dependent. Additionally, the structures, systems, and/or devices described herein may be embodied as integrated components or as separate components. For purposes of comparing various embodiments, certain aspects and advantages of these embodiments are described. Not necessarily all such aspects or advantages are achieved by any particular embodiment. Thus, for example, various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may also be taught or suggested herein.

(24) Certain standard anatomical terms of location are used herein to refer to the anatomy of animals, and namely humans, with respect to the preferred embodiments. Although certain spatially relative terms, such as “outer,” “inner,” “upper,” “lower,” “below,” “above,” “vertical,” “horizontal,” “top,” “bottom,” and similar terms, are used herein to describe a spatial relationship of one device/element or anatomical structure to another device/element or anatomical structure, it is understood that these terms are used herein for ease of description to describe the positional relationship between element(s)/structure(s), as illustrated in the drawings. It should be understood that spatially relative terms are intended to encompass different orientations of the element(s)/structure(s), in use or operation, in addition to the orientations depicted in the

drawings. For example, an element/structure described as “above” another element/structure may represent a position that is below or beside such other element/structure with respect to alternate orientations of the subject patient or element/structure, and vice-versa.

(25) Traditional puncture needles used to form openings at target tissue sites in minimally invasive transcatheter approaches can often include a puncture component with a pointed sharp distal end. The target tissue sites can be on a tissue of any number of internal vessels, channels, chambers, and/or organs. Advancing a puncture component with a pointed sharp distal end through a lumen of a delivery catheter positioned in a tortuous anatomical pathway can result in the puncture needle becoming jammed within the delivery catheter lumen. The pointed sharp distal end can contact and become lodged within a wall of the lumen as the puncture component is inserted through the lumen, including through a bend in the delivery catheter lumen. The pointed sharp distal end can prick a portion of the lumen wall, and subsequently become stuck to the lumen wall, thereby preventing insertion of the puncture needle through the delivery catheter to the target tissue site. Traditional puncture needles may not be insertable to a puncture needle outlet opening of a lumen of a delivery catheter already positioned within a tortuous anatomical pathway. For example, traditional puncture needles may not be advanceable from a proximal portion, including a proximal end, to an outlet opening on a distal portion of the delivery catheter lumen without the puncture needles becoming jammed within the delivery catheter lumen. Curves along the anatomical pathway can result in bends in the delivery catheter, and thereby in the delivery catheter lumen, preventing delivery of traditional puncture needles to the target site. A traditional puncture needle may need to be pre-loaded within a delivery catheter lumen such that the puncture needle only needs to be advanced a short distance within a distal portion of the delivery catheter lumen to reach the puncture needle outlet opening positioned at or proximate to a target site. A need to pre-load the puncture needle in the delivery catheter lumen can prevent use of the lumen for other purposes, for example preventing exchange of the puncture needle for other instrumentation, thereby requiring use of a delivery catheter with an increased profile.

(26) Puncture needles described herein can be navigated through a lumen of a delivery catheter positioned in a tortuous anatomical pathway without the puncture needles becoming jammed within the delivery catheter lumen. A puncture needle as described herein can be advanced from a proximal portion, such as a proximal end, to a puncture needle outlet opening on a distal portion of the delivery catheter lumen without the puncture needle becoming stuck within the delivery catheter lumen. The puncture needle can be advanced through one or more bends in the delivery catheter lumen without having any portion of the puncture needle becoming lodged within a delivery catheter lumen wall. The delivery catheter can be first positioned within an anatomical pathway to place the puncture needle outlet opening adjacent or proximate to a target tissue site, and the puncture needle can be subsequently inserted through the already positioned delivery catheter to the puncture needle outlet opening. The delivery catheter does not need to be pre-loaded with the puncture needle prior to positioning the delivery catheter within the anatomical pathway. The puncture needle can be inserted and/or retracted through a delivery catheter already positioned within a patient, enabling exchange of the puncture needle for one or more other instrumentations. Such puncture needles can thereby enable use of delivery catheters with a reduced profile, such as compared to delivery catheters used to accommodate pre-loaded punctured needles.

(27) A puncture needle configured to be advanced through one or more bends of a delivery catheter lumen can include a spacer associated with an elongate portion of the puncture needle, and/or a puncture component which includes a distal edge with a blade edge thereon. The puncture component can be configured to pierce the tissue and can be associated with a distal end of the puncture needle elongate portion. The spacer can be on a distal portion of the elongate portion adjacent to or a predetermined distance proximal of the puncture component. The spacer can comprise at least a portion which extends laterally from the distal portion. The spacer can be configured to contact a wall of the delivery catheter lumen to prevent contact between the puncture

component and the wall when the puncture component is advanced through a bend in the delivery catheter lumen. The spacer can be configured to contact the delivery catheter lumen wall so as to provide sufficient distance between a distal end of the puncture component and the lumen wall at and/or proximate to a bend in the lumen such that the distal end cannot contact the lumen wall at and/or proximate to the bend. The spacer can prevent any portion of the puncture component from becoming lodged within the lumen wall, facilitating insertion of the puncture needle through bends in the delivery catheter lumen. A puncture needle comprising a spacer and a puncture component with a pointed sharp distal end can be advanced through one or more bends of a delivery catheter lumen without the puncture needle becoming jammed in the lumen. Avoiding contact between the pointed sharp distal end and the lumen wall can also prevent damage to the lumen wall and/or damage to the pointed sharp distal end to thereby prevent the pointed sharp distal end from becoming unsuitable for piercing tissue to form an opening at the target tissue site.

(28) In some embodiments, the spacer can extend radially from the distal portion the elongate portion. In some embodiments, the spacer can be circumferentially positioned on the distal portion. For example, the spacer can have a ring shape. In some embodiments, the spacer can be partially circumferentially positioned on the distal portion. In some embodiments, the spacer can comprise a plurality of distinct portions arranged around the distal portion. The spacer can comprise a plurality of separate panels. The spacer can comprise a plurality of distinct lateral surface portions. In some embodiments, the spacer can comprise a plurality of distinct portions arranged partially circumferentially around the distal portion.

(29) The spacer may be symmetrical around an axis collinear with a longitudinal axis of the puncture needle. In some embodiments, the spacer is asymmetrical around the axis. A lateral surface of the spacer can be parallel or substantially parallel to a longitudinal axis of the puncture needle, curve and/or bow outwardly, and/or curve and/or bow inwardly. For example, the lateral surface may comprise a convex curvature and/or a concave curvature. In some embodiments, the spacer can comprise a plurality of distinct lateral surface portions which curve and/or bow outwardly. In some embodiments, a shape of the spacer can be predetermined based at least in part on an anticipated direction and/or anticipated degree of curvature of one or more bends in the delivery catheter lumen, including an anticipated direction and/or degree of a sharpest bend. In some embodiments, a shape of a spacer can be selected to facilitate insertion and/or retraction of the spacer through the delivery catheter lumen, for example to reduce friction between the spacer the delivery catheter lumen wall.

(30) In some embodiments, a width and/or a length of the spacer can be selected based on at least a distance between the spacer and a distal end of the puncture component. For example, a spacer at a predetermined distance proximal of the puncture component can comprise a width wider, and/or a length shorter, than that of a spacer adjacent to the puncture component. In some embodiments, at least a portion of the spacer can be inserted into an opening formed at the target tissue site to serve as a dilator to enlarge the opening. The spacer can comprise a width selected based at least in part on a desired enlargement of the opening.

(31) In some embodiments, a puncture needle configured to be advanceable through one or more bends of a delivery catheter lumen can comprise a puncture component with a blade edge on a distal edge. The blade edge can be sharp so as to be able to pierce tissue to form an opening at the target tissue site. In some embodiments, the blade edge can be a curved blade edge, for example a convexly curved blade edge. In some embodiments, the blade edge can comprise a linear portion between opposing curved portions. For example, the blade edge can be a linear portion between rounded corners. The curvature in the blade edge can prevent the blade edge from becoming stuck against the delivery catheter lumen wall. For example, the curved blade edge and/or the blade edge comprising the linear portion between opposing curved corners may contact the delivery catheter lumen wall at a bend in the delivery catheter lumen and glide along the wall without sticking to the wall. A puncture component with a blade edge on a distal edge can comprise a variety of shapes. In

some embodiments, the puncture component comprising the blade edge can comprise a rounded paddle shape, the blade edge being on a curved distal edge of the rounded paddle.

(32) Puncture needles described herein can be used to form openings at various target tissue locations within the heart to enable delivery of medical implant devices and/or therapies to the heart. The medical implant devices and/or therapies can be delivered to ameliorate various heart abnormalities. In some embodiments, the puncture needles can be used to form openings on the left atrial wall. For example, delivery of implant devices and/or therapies to the heart can be performed for treatment of elevated pressure in the left atrium. The puncture needles can be used to form openings on portions of the left atrial wall accessible from within the coronary sinus. FIG. 1 shows examples of pathways to the coronary sinus, such that puncture needles can be positioned within the coronary sinus to access the wall of the left atrium (LA). Openings can be formed on the left atrial wall for delivery of any number of medical therapies and/or medical devices, including deployment of expandable shunts into the left atrial wall. As shown in FIG. 1, access into the coronary sinus (CS) can be made through the right atrium (RA) via the coronary sinus ostium (CS ostium). The right atrium can be accessed via the superior vena cava (SVC) or via the inferior vena cava (IVC). For example, a transjugular or trans-subclavian approach can be used to access the right atrium via the superior vena cava. A delivery catheter for delivering a puncture needle can be inserted into the subclavian veins or jugular veins, and advanced into the superior vena cava (e.g., along pathway A). Alternatively, a transfemoral approach can be used to position a delivery catheter into the inferior vena cava (e.g., along pathway B). Other access routes may also be used for insertion of delivery catheters into the vasculature.

(33) Reference herein to “catheters” and/or “delivery catheters” can refer or apply generally to any type of elongate tubular delivery device comprising an inner lumen configured to slidably receive instrumentation, such as for positioning within an atrium or coronary sinus, including for example delivery sheaths and/or cannulas.

(34) The term “associated with” is used herein according to its broad and ordinary meaning. For example, where a first feature, element, component, device, or member is described as being “associated with” a second feature, element, component, device, or member, such description should be understood as indicating that the first feature, element, component, device, or member is physically coupled, attached, or connected to, integrated with, embedded at least partially within, or otherwise physically related to the second feature, element, component, device, or member, whether directly or indirectly.

(35) The term “lumen” as used herein can refer to a tubular cavity and/or space, and/or one or more structures which, form, surround and/or define the tubular cavity and/or space, including one or more surfaces which form, surround and/or define the tubular space and/or cavity.

(36) FIGS. 2 through 10 show various examples of puncture needles and associated spacers. In some embodiments, a spacer can be integrally formed with the associated puncture needle, such as with an elongate portion of the puncture needle. In some embodiments, the spacer can be separable from the elongate portion. For example, the spacer can be a distinct and separate component where the spacer is configured to be coupled to the elongate portion. The spacer can be selected for coupling to the elongate portion. Additionally, although the spacer is described as being associated with the elongate portion of the puncture needle, in some embodiments, the spacer can instead be associated with a distal exterior portion of a puncture needle sheath. The puncture needle sheath can be configured to receive the puncture needle and be advanced through a delivery catheter lumen together with the puncture needle to or proximate to a target tissue site. The spacer can be on the distal exterior portion of the puncture needle sheath. In some embodiments, the puncture needle can be advanced relative to the puncture needle sheath such that the puncture needle can be used to pierce the target tissue. In some embodiments, the puncture needle sheath can be retracted relative to the puncture needle after the puncture needle sheath and the puncture needle have been inserted to the target tissue site, so as to facilitate deployment of the puncture needle for piercing the tissue.

(37) The spacers can be implemented in any number of suitable manners. In some embodiments, a spacer can be pre-formed to include one or more configurations as described herein. In some embodiments, a spacer can be configured to transform from a collapsed configuration into a deployed configuration in response to a trigger and/or actuation, the deployed configuration comprising one or more configurations as described herein. In some embodiments, a spacer can comprise an inflatable member, for example, being configured to be inflated prior to or after insertion thereof into an anatomical pathway. The inflatable member can be inflated such that the spacer can assume one or more configurations as described herein.

(38) Although the puncture component as described with reference to FIGS. 2 through 10 have a conical configuration, it will be understood that the puncture component can comprise a number of other configurations, including a pre-curved puncture component with a pointed sharp distal end.

(39) FIG. 2 is a side view of an example of a puncture needle 100 and an associated spacer 150 positioned within a lumen 52 of a delivery catheter 50. A cross-sectional view of the delivery catheter 50 is shown. The puncture needle 100 can comprise an elongate portion 102 and the spacer 150 can be on a distal portion 104 of the elongate portion 102. The puncture needle 100 can comprise a puncture component 106 at a distal end (not shown) of the elongate portion 102. The puncture component 106 can comprise a proximal end 108 and a pointed sharp distal end 110. For example, the puncture component 106 can comprise a conical shape.

(40) The spacer 150 can extend radially from the elongate portion and can be configured to contact a wall 54 of the lumen 52 while the puncture needle 100 is advanced through the lumen 52 to prevent contact between the wall 54 of the lumen 52 and the pointed sharp distal end 110 of the puncture component 106. Contact between the pointed sharp distal end 110 and the wall 54 can result in the pointed sharp distal end 110 becoming lodged within the wall 54 and/or stuck to the wall 54, causing the puncture needle 100 to become jammed within the lumen 52. Contact between the pointed sharp distal end 110 and the wall 54 can damage the delivery catheter 50 and/or the pointed sharp distal end 110. The spacer 150 can contact the wall 54 such that sufficient distance is maintained between the pointed sharp distal end 110 and the wall 54 while the puncture needle 100 is advanced through the delivery catheter lumen 52, including while the puncture needle 100 is advanced through a bend in the lumen 52, to prevent contact between the pointed sharp distal end 110 and the wall 54. The puncture needle 100 can thereby be inserted and/or retracted through the delivery catheter lumen 52 without being jammed within the lumen 52, allowing the puncture needle 100 to be exchangeable with one or more other surgical instruments, including instrumentations for delivering one or more therapies and/or one or more medical devices to a target tissue site.

(41) The spacer 150 can have a proximal end 152 and a distal end 154, and a lateral portion 156 extending between the proximal end 152 and the distal end 154. The spacer 150 can have a proximal edge portion 158, such as an edge along which the proximal end 152 meets the lateral portion 156, and a distal edge portion 160, such as an edge along which the distal end 154 meets the lateral portion 156. In the example shown in FIG. 2, the puncture needle 100 is positioned in a bend of the delivery catheter lumen 52. In some embodiments, to prevent contact between the pointed sharp distal end 110 and the wall 54, the lateral portion 156 of the spacer 150 can contact the wall 54 of the delivery catheter lumen 52 along an inner edge of the radius of curvature. In some embodiments, the proximal end 152 and/or the proximal edge portion 158, and the distal end 154 and/or the distal edge portion 160, can contact the wall 54 along an outer edge of the radius of curvature of the bend. Contact between the wall 54 and the lateral portion 156, the proximal end 152, the proximal edge portion 158, the distal end 154, and/or the distal edge portion 160 can provide sufficient space between the pointed sharp distal end 110 and the wall 54 such that the pointed sharp distal end 110 does not prick the wall 54 while the puncture needle 100 is advanced through the bend.

(42) In some embodiments, the spacer 150 can be adjacent to the puncture component 106. For

example, the proximal end **108** of the puncture component **106** can be adjacent to the distal end **154** of the spacer **150**. In some embodiments, the distal end **154** of the spacer **150** can be adjacent to the distal end of the elongate portion **102**. The distal end **154** of the spacer **150** can be adjacent to the proximal end **108** of the puncture component **106**. Alternatively, the spacer **150** can be at a predetermined distance proximal of the puncture component **106**. For example, the distal end **154** of the spacer **150** can be a predetermined distance proximal of the proximal end **108** of the puncture component **106**. In some embodiments, the distal end **154** of the spacer **150** can be at a predetermined distance proximal of the distal end of the elongate portion **102**.

(43) FIG. 3A is a perspective view of the puncture needle **100** and the spacer **150**. The spacer **150** can be circumferentially positioned around the distal portion **104** of the elongate portion **102**. For example, the spacer **150** can have a ring shape. In some embodiments, the spacer **150** can be coaxial with a longitudinal axis L.sub.100 of the puncture needle **100**. The spacer **150** can be symmetrical around an axis which is collinear with a longitudinal axis L.sub.100 of the puncture needle **100**. The longitudinal axis L.sub.100 can extend centrally along a length of the elongate portion **102**, such as through a proximal portion (not shown) and the distal portion **104** of the elongate portion **102**.

(44) FIG. 3B shows a longitudinal cross-sectional view of the spacer **150**. The longitudinal cross section can be taken along a longitudinal cross-sectional plane P.sub.150. The longitudinal cross-sectional plane P.sub.150 can extend laterally from the distal portion **104** along the longitudinal axis L.sub.100 of the puncture needle **100**. For example, a longitudinal cross section of the spacer **150** can comprise a rectangular or substantially rectangular shape. The lateral portion **156** can comprise a lateral surface **162** parallel or substantially parallel to the longitudinal axis L.sub.100 of the puncture needle **100**.

(45) A width and/or a length of the spacer **150** can be predetermined to prevent contact between the pointed sharp distal end **110** and the delivery catheter lumen wall **54**. The length can be a longitudinal dimension parallel or substantially parallel to the longitudinal axis L.sub.100. For example, the length can be a dimension extending between the proximal end **152** and the distal end **154** of the spacer **150**. The width can be a lateral dimension which is perpendicularly or substantially perpendicularly to the length, such as perpendicular or substantially perpendicular to the longitudinal axis L.sub.100 of the puncture needle **100**. For example, the width can extend between opposing portions of the lateral surface **162**.

(46) As described herein, the spacer **150** can be adjacent to, or alternatively, a predetermined distance proximal of, the puncture component **106**. In some embodiments, the width and/or the length of the spacer **150** can be predetermined based at least in part on a distance between the spacer **150** and the pointed sharp distal end **110** of the puncture component **106**. In some embodiments, the width can be directly related to the distance between the spacer **150**, such as the distal end **154**, and the pointed sharp distal end **110**. For example, a wider width can be selected for a spacer positioned at a distance proximal of the puncture component **106** than for a spacer positioned adjacent to the puncture component **106**. In some embodiments, the length of the spacer **150** can be inversely related to a distance between the spacer **150**, such as the distal end **154**, and the pointed sharp distal end **110**. For example, a shorter length can be selected for a spacer positioned at a distance proximal of the puncture component **106** than for a spacer positioned adjacent to the puncture component **106**. In some embodiments, the length does not depend on the distance between the spacer **150** and the pointed sharp distal end **110**.

(47) In some embodiments, the proximal end **152** and/or the distal end **154** can be a planar or substantially planar surface. Alternatively, the spacer **150** can comprise a proximal taper and/or a distal taper, such as a taper from the lateral portion **156** toward the proximal end **152** and/or the distal end **154**, respectively. The taper may include a curved and/or linear taper. In some embodiments, the spacer **150** can comprise a proximal taper and/or a distal taper to facilitate retraction and/or insertion of the spacer **150** through the delivery catheter lumen **52**.

(48) In some embodiments, the spacer **150** can serve as a dilator configured to be inserted into the opening formed at a target tissue site by the puncture component **106** so as to enlarge the opening. The puncture needle **100** can be advanced further after the opening is formed such that at least a portion of the spacer **150** is inserted into the opening. In some embodiments, the width of the spacer **150** can be selected based at least in part on a desired dilation of the opening. In some embodiments, the width of the spacer **150** can be selected so as not to be too large to impede insertion of the spacer **150** into the opening. In some embodiments, the spacer **150** can comprise a distal taper to facilitate insertion of the spacer **150** into the opening. In some embodiments, the spacer **150** is not configured to be inserted into the opening. In some embodiments, the distal end **154** of the spacer **150** can be planar or substantially planar to prevent or reduce insertion of the spacer **150** into the opening.

(49) FIGS. **4** through **10** show additional examples of spacers. FIG. **4A** is a perspective view of an example of a puncture needle **200** and an associated spacer **250** comprising an outward-curving lateral surface **262**. FIG. **4B** shows a longitudinal cross-sectional view of the spacer **250**. Referring to FIG. **4A**, the spacer **250** can be circumferentially positioned around an elongate portion **202** of the puncture needle **200**. For example, the spacer **250** can extend radially from a distal portion **204** of the elongate portion **202**. The spacer **250** can have a ring shape. In some embodiments, the spacer **250** can be coaxial with the puncture needle **200**. The spacer **250** can be symmetrical around an axis collinear with the longitudinal axis L.sub.200 of the puncture needle **200**. The spacer **250** can have a proximal end **252** and a distal end **254**, and a lateral portion **256** between the proximal end **252** and the distal end **254**. The lateral portion **256** can comprise a lateral surface **262** which comprises a convex curvature. For example, the lateral surface **262** can curve and/or bow outward. Opposing portions of the lateral portion **256**, such as opposing portions of the lateral surface **262**, can contact a delivery catheter lumen wall along an inner edge and an outer edge of a curvature in the delivery catheter lumen.

(50) Referring to FIG. **4B**, a longitudinal cross section of the spacer **250** along the longitudinal cross-sectional plane P.sub.250 can comprise a segment of a circle. For example, the convex curvature can have a uniform or substantially uniform radius of curvature. In some embodiments, the longitudinal cross section along the longitudinal cross-sectional plane P.sub.250 can comprise a segment of an oval. The longitudinal cross-sectional plane P.sub.250 can extend laterally from the distal portion **204** of the elongate portion **202** along the longitudinal axis L.sub.200. The lateral portion **256** comprising the outward curvature can reduce contact, thereby reducing friction, between the spacer **250** and a delivery catheter lumen wall to facilitate insertion and/or retraction of the puncture needle **200** through the delivery catheter lumen.

(51) FIG. **5A** is a perspective view of an example of a puncture needle **300** and an associated spacer **350** comprising an inward curving lateral surface **362**. FIG. **5B** shows a longitudinal cross-sectional view of the spacer **350**. Referring to FIG. **5A**, the spacer **350** can be circumferentially positioned around an elongate portion **302** of the puncture needle **300**, such as around a distal portion **304** of the elongate portion **302**. The spacer **350** can have a ring shape. In some embodiments, the spacer **350** can be coaxial with the puncture needle **300**. The spacer **350** can be symmetrical around an axis which is collinear with a longitudinal axis L.sub.300 of the puncture needle **300**. The spacer **350** can have a proximal end **352**, a distal end **354**, and a lateral portion **356** between the proximal end **352** and the distal end **354**. The lateral portion **356** can have a lateral surface **362**. The lateral surface **362** can comprise a concave curvature. For example, the lateral surface **362** can curve or bow inward.

(52) Referring to FIG. **5B**, a longitudinal cross-sectional view of the spacer **350** is shown. The longitudinal cross section of the spacer **350** can be taken along the longitudinal cross-sectional plane P.sub.350. The longitudinal cross-sectional plane P.sub.350 can extend laterally from the distal portion **304** of the elongate portion **302** along the longitudinal axis L.sub.300. The longitudinal cross section of the lateral portion **356** can comprise a concave curvature, the lateral

surface **362** curving inward. In some embodiments, the concave curvature can have a uniform radius of curvature. For example, the concave curvature can comprise a segment of a circle. In some embodiments, the radius of curvature of the concave curvature can be non-uniform. For example, the concave curvature can comprise a segment of an oval.

(53) The spacer **350** can have a proximal edge portion **358**, such as an edge along which the proximal end **352** meets the lateral portion **356**, and a distal edge portion **360**, such as an edge along which the distal end **354** meets the lateral portion **356**. In some embodiments, when the puncture needle **300** is advanced through a bend in a delivery catheter lumen, the lateral portion **356** can contact a wall of the delivery catheter lumen along an inner edge of the curvature in the bend. In some embodiments, the proximal end **352** and/or the proximal edge portion **358**, and the distal end **354** and/or the distal edge portion **360**, can contact the wall along on an outer edge of the curvature in the delivery catheter lumen. In some embodiments, the concave curvature of the lateral portion **356** can provide reduced contact between the spacer **350** and the delivery catheter lumen wall, thereby reducing friction between the spacer **350** and the delivery catheter lumen wall so as to facilitate insertion and/or retraction of the puncture needle **300**.

(54) FIG. **6A** is a perspective view of an example of a puncture needle **400** and an associated spacer **450**, where the spacer **450** includes an outwardly curving lateral surface **462**, a distal portion of the lateral surface **462** having a smaller radius of curvature than a proximal portion of the lateral surface **462**. FIG. **6B** shows a longitudinal cross-sectional view of the spacer **450**. Referring to FIG. **6A**, the spacer **450** can be circumferentially positioned around an elongate portion **402** of the puncture needle **400**, such as a distal portion **404** of the elongate portion **402**. In some embodiments, the spacer **450** can be coaxial with the puncture needle **400**. The spacer **450** can be symmetrical around an axis collinear with a longitudinal axis L.sub.400 of the puncture needle **400**. The spacer **450** can have a proximal end **452**, a distal end **454**, and a lateral portion **456** between the proximal end **452** and the distal end **454**. The lateral portion **456** can have a lateral surface **462**. The lateral portion **456** can comprise an outward bow and/or curvature with a non-uniform radius of curvature. For example, the lateral surface **462** can comprise a convex curvature with a non-uniform radius of curvature, where the radius of curvature increases along a direction extending from the distal end **454** to the proximal end **452**. A distal portion **460** of the lateral surface **462** can have a smaller radius of curvature than a proximal portion **458** of the lateral surface **462**.

(55) Referring to FIG. **6B**, a longitudinal cross section of the spacer **450** taken along the longitudinal cross-sectional plane P.sub.450 is shown. The longitudinal cross-sectional plane P.sub.450 can extend laterally from the distal portion **404** along the longitudinal axis L.sub.400. As shown in the longitudinal cross section of the spacer **450**, the lateral surface **462** can comprise a convex curvature, where the radius of curvature increases along a direction extending from the distal end **454** toward the proximal end **452**. A width of the spacer **450** can generally decrease along a direction extending from the distal end **454** to the proximal end **452**. The width of the spacer **450** can be a dimension which extends between opposing portions of the lateral surface **462**. In some embodiments, the longitudinal cross section of the spacer **450** can comprise a segment of a teardrop shape, where a wider portion of the segment is oriented toward the puncture component **406** of the puncture needle **400**. When the puncture needle **400** is inserted through a bend in a delivery catheter lumen, opposing portions of the lateral surface **462** can contact a wall of the delivery catheter lumen along an inner edge and an outer edge of the bend. For example, opposing portions of the distal portion **460** of the lateral surface **462** can contact the delivery catheter lumen wall along the inner edge and outer edge of the bend.

(56) FIG. **7A** is a perspective view of an example of a puncture needle **500** and an associated spacer **550**, where the spacer **550** includes an outwardly curving lateral surface **562**, a proximal portion **558** of the lateral surface **562** having a smaller radius of curvature than a distal portion **560** of the lateral surface **562**. FIG. **7B** shows a longitudinal cross-sectional view of the spacer **550**. The spacer **550** can have an orientation opposite that of the spacer **450** described with reference to

FIGS. 6A and 6B. The spacer **550** can comprise other features which are the same as or similar to that of the spacer **450**. For example, referring to FIG. 7A, the spacer **550** can be circumferentially positioned around a distal portion **504** of the elongate portion **502**. In some embodiments, the spacer **550** can be coaxial with the puncture needle **500**. The spacer **550** can have symmetry around an axis collinear with a longitudinal axis L.sub.500 of the puncture needle **500**. Referring to FIG. 7B, a longitudinal cross section of the spacer **550** taken along the longitudinal cross-sectional plane P.sub.550 shows that the lateral surface **562** of a lateral portion **556** extending between a proximal end **552** and a distal end **554** can have a convex curvature. The convex curvature can have a radius of curvature which decreases along a direction extending from the distal end **554** toward the proximal end **552**. The longitudinal cross-sectional plane P.sub.550 can extend laterally from the distal portion **504** along the longitudinal axis L.sub.500. A width of the spacer **550**, such as a dimension which extends between opposing portions of the lateral surface **562**, can generally increase along a direction extending from the distal end **554** to the proximal end **552**. In some embodiments, the longitudinal cross section of the spacer **550** can comprise a segment of a teardrop shape, where a wider portion of the segment is oriented away from the puncture component **506** of the puncture needle **500**. When the puncture needle **500** is inserted through a bend in a delivery catheter lumen, opposing portions of the lateral surface **562** can contact a wall of the delivery catheter lumen along an inner edge and an outer edge of the bend. For example, opposing portions of the proximal portion **558** of the lateral surface **562** can contact the delivery catheter lumen wall along the inner edge and outer edge of the bend.

(57) FIG. 8A is a perspective view of an example of a puncture needle **600** and an associated spacer **650** which is asymmetrical around an axis collinear with a longitudinal axis L.sub.600 of the puncture needle **600**. The puncture needle **600** can have a puncture component **606** at a distal end of an elongate portion. The spacer **650** can be coaxial with the puncture needle **600**. FIG. 8B is a longitudinal cross-sectional view of the puncture needle **600**. Referring to FIG. 8A, the spacer **650** can be circumferentially positioned around a distal portion **604** of the elongate portion **602**. The spacer **650** can have a proximal end **652**, a distal end **654**, and a lateral portion **656** between the proximal end **652** and the distal end **654**. The lateral portion **656** can have a lateral surface **662**. The lateral surface **662** and the proximal end **652** and distal end **654** can meet to form a proximal edge portion **658** and a distal edge portion **660**, respectively. The lateral portion **656** can comprise both an outward curvature and an inward curvature. For example, opposing portions of the lateral surface **662** can comprise a convex curvature and a concave curvature such that the spacer **650** is asymmetrical around the longitudinal axis L.sub.600.

(58) Referring to FIG. 8B, a longitudinal cross-sectional view of the spacer **650** is shown. A longitudinal cross section of the spacer **650** along the longitudinal cross-sectional plane P.sub.650 is shown. The longitudinal cross-sectional plane P.sub.650 can extend laterally from the distal portion **604** in opposite directions along the longitudinal axis L.sub.600. The longitudinal cross section of the spacer **650** taken along the cross-sectional plane P.sub.650 shows that the lateral surface **662** can comprise both a convex curvature and a concave curvature. For example, a first portion of the lateral surface **662** can comprise the convex curvature and a portion of the lateral surface **662** opposite the first portion can comprise the concave curvature. In some embodiments, the radius of curvature of the convex curve and/or the concave curve can be uniform or substantially uniform. In some embodiments, the radius of curvature of the convex curve and/or the concave curve can be non-uniform.

(59) In some embodiments, the orientation of the convex curvature and the concave curvature can depend on an anticipated curvature of a known bend in the anatomical pathway into which a delivery catheter is advanced. For example, the orientation can be selected based on a direction of an anticipated sharpest bend in the anatomical pathway, and thereby based on a direction of an anticipated sharpest bend in the delivery catheter lumen. In some embodiments, the concave portion of the lateral surface **662** can be configured to be oriented toward an outer edge of the bend

and the convex portion of the lateral surface **662** can be oriented toward an inner edge of the bend. In some embodiments, the convex portion of the lateral surface **662** can contact a delivery catheter lumen wall along the inner edge of the bend. In some embodiments, the proximal end **652** and/or the proximal edge portion **658**, and the distal end **654** and/or the distal edge portion **660**, can contact the delivery catheter lumen wall along the outer edge.

(60) In some embodiments, a spacer asymmetrical around an axis collinear with a longitudinal axis of the puncture needle can comprise another configuration. For example, opposing lateral surface portions can comprise a curved portion, such as a convex and/or concave curvature, and a portion parallel or substantially parallel to the longitudinal axis.

(61) FIGS. **9A** and **9B** show an example of a puncture needle **700** and an associated spacer **750** which includes a pair of panels positioned in an opposing manner on an elongate portion **702** of the puncture needle **700**. The puncture needle **700** can have a puncture component **706** at a distal end of the elongate portion **702**. FIG. **9A** is a perspective view, while FIG. **9B** is a longitudinal cross-sectional view. The spacer **750** can include a plurality of discrete panels arranged around a distal portion **704** of the elongate portion **702**. For example, the spacer **750** can include a first panel **750a** and a second panel **750b** arranged around the distal portion **704** of the elongate portion **702**. The first panel **750a** and the second panel **750b** can be arranged in opposing fashion around the distal portion **704**. The panels **750a**, **750b** can be partially circumferentially arranged around the distal portion **704**. For example, the panels **750a**, **750b** can each comprise an arcuate shape and can be spaced from one another by a pair of opposing gaps (only one of the gaps is visible in FIG. **9A**). The panels **750a**, **750b** can be arranged to surround a corresponding portion of the elongate portion **702**. In some embodiments, the first panel **750a** and the second panel **750b** can have the same or similar configurations. In some embodiments, the separation between the panels can provide improved flexibility to facilitate insertion of the spacer **750** through a delivery catheter lumen placed within tortuous anatomical pathways. The gaps separating the first panel **750a** and the second panel **750b** can be predetermined to provide improved flexibility while allowing the spacer **750** to maintain a desired distance between the puncture component **706** and the delivery catheter lumen wall.

(62) The first panel **750a** and the second panel **750b** can each comprise a respective proximal end **752a**, **752b**, a respective distal end **754a**, **754b**, and a respective lateral portion **756a**, **756b**. Each lateral portion **756a**, **756b** can comprise a respective lateral surface **762a**, **762b**. The first panel **750a** can comprise a proximal edge portion **758a**, such as an edge along which the proximal end **752a** meets the lateral portion **756a**, and a distal edge portion **760a**, such as an edge along which the distal end **754a** meets the lateral portion **756a**. The second panel **750b** can comprise a second proximal edge portion **758b**, such as an edge along which the second proximal end **752b** meets the second lateral portion **756b**, and a second distal edge portion **760b**, such as an edge along which the second distal end **754b** meets the second lateral portion **756b**. For example, as the puncture needle **700** is advanced through a bend in a delivery catheter lumen, the first lateral surface **762a** can contact a lumen wall along an inner edge of the bend. In some embodiments, the second proximal end **752b** and/or the second proximal edge portion **758b**, and the second distal end **754b** and/or the second distal edge portion **760b**, can contact the lumen wall along an outer edge of the bend. In some embodiments, the second lateral surface **762b** can contact a lumen wall along an inner edge of the bend, and the first proximal end **752a** and/or the first proximal edge portion **758a**, and the first distal end **754a** and/or the first distal edge portion **760a**, can contact the lumen wall along an outer edge of the bend.

(63) The cross-sectional view of FIG. **9B** is taken along the longitudinal cross-sectional plane P.sub.750. The longitudinal cross-sectional plane P.sub.750 can extend laterally from the distal portion **704** of the elongate portion **702** along the longitudinal axis L.sub.700. The longitudinal cross section of the first panel **750a** and the second panel **750b** are shown. In some embodiments, a connector member **770a**, **770b** can couple the first panel **750a** and the second panel **750b** to the

distal portion **704** of the elongate portion **702**. The lateral portions **756a**, **756b**, of the first panel **750a** and the second panel **750b** can be parallel or substantially parallel to the elongate portion **702**. (64) It will be understood that the number and/or shape of the panels **750a**, **750b** are used for illustrative purposes. In some embodiments, a spacer can have more than two panels arranged around an elongate portion of a puncture needle. In some embodiments, one or more of the panels can comprise a different shape. For example, one or more of the panels can comprise one or more shapes as described herein, including a panel comprising an outward curvature and/or an inward curvature. A lateral surface of the panel can comprise a convex curvature and/or a concave curvature, where the curvature may or may not have a uniform radius of curvature.

(65) FIGS. **10A**, **10B** and **10C** show an example of a puncture needle **800** and an associated spacer **850** comprising a plurality of distinct outwardly curving portions circumferentially positioned around an elongate portion **802** of the puncture needle **800**. The puncture needle **800** can have a puncture component **806** at a distal end of the elongate portion **802**. FIG. **10A** is a perspective view of the puncture needle **800** and FIG. **10B** is a view of the puncture needle **800** from a distal end. Referring to FIG. **10A**, the spacer **850** can comprise four portions **850a**, **850b**, **850c** (not shown), **850d** positioned around a distal portion **804** of the elongate portion **802**. In some embodiments, the four portions **850a**, **850b**, **850c**, **850d** can be four uniform or substantially uniform portions evenly distributed around the distal portion **804**, each of the four portions **850a**, **850b**, **850c**, **850d** comprising an outward bow. The four portions **850a**, **850b**, **850c**, **850d** can be four distinct, and uniform or substantially uniform, lobes evenly distributed around the circumference of the distal portion **804**. The four portions **850a**, **850b**, **850c**, **850d** are shown in FIG. **10B**.

(66) The spacer **850** can have a proximal end **852** and a distal end **854**. Each of the four portions **850a**, **850b**, **850c**, **850d** can have a corresponding lateral portion **856a**, **856b**, **856c**, **856d** between the proximal end **852** and the distal end **854**. Each lateral portion **856a**, **856b**, **856c**, **856d** can comprise a corresponding lateral surface **862a**, **862b**, **862c**, **862d**. In some embodiments, each of the lateral portions **856a**, **856b**, **856c**, **856d** can comprise an outward curvature, such as a convex curvature. For example, each of the four distinct lateral surfaces **862a**, **862b**, **862c**, **862d** can comprise a convex curvature. The lateral portions **856a**, **856b**, **856c**, **856d** can each comprise a convex curvature along both a longitudinal axis and a lateral axis, where the lateral axis is perpendicular or substantially perpendicular to the longitudinal axis. As described in further detail herein, the longitudinal axis can extend along a direction parallel or substantially parallel with a longitudinal axis of the puncture needle **800**, L.sub.800. For example, the longitudinal axis can be collinear with L.sub.800. In some embodiments, a lateral cross section of each lateral portion **856a**, **856b**, **856c**, **856d** along a plane extending along the lateral axis can comprise a convex curvature. The radius of curvature of the convex curve may or may not be uniform. In some embodiments, the lateral cross section can comprise a segment of a circle. In some embodiments, the lateral cross section can comprise a segment of an oval.

(67) In FIG. **10C**, a longitudinal cross-sectional view of the spacer **850** is shown. Specifically, a longitudinal cross-sectional view of the spacer **850** along the longitudinal cross-sectional plane P.sub.850 is shown. The longitudinal cross-sectional plane P.sub.850 can extend laterally from the distal portion **804** of the elongate portion **802** along the longitudinal axis L.sub.800. In some embodiments, a longitudinal cross section of the spacer **850** can comprise a segment of a circle. In some embodiments, a longitudinal cross section of the spacer **850** can comprise a segment of an oval. For example, a longitudinal cross section of the lateral portions **856a**, **856b**, **856c**, **856d** can comprise a segment of a circle or a segment of an oval. The lateral surfaces **862a**, **862b**, **862c**, **862d** can comprise a convex curvature. The radius of curvature of the curvature may or may not be uniform.

(68) In some embodiments, a spacer can comprise more or fewer distinct lateral surface portions. One or more lateral surface portions of the plurality of distinct lateral surface portion can curve and/or bow inwardly, for example comprising a concave curvature. The curvature in the one or

more lateral surface portions can reduce friction between the spacer and a delivery catheter lumen wall so as to facilitate insertion and/or retraction of the spacer. One or more lateral surface portions of the plurality of distinct lateral surface portion can be parallel or substantially parallel to a longitudinal axis of an associated puncture needle. In some embodiments, the spacer can be asymmetrical around an axis coaxial with the longitudinal axis of the puncture needle. In some embodiments, a shape of the spacer can be predetermined based on an anticipated direction and/or degree of one or more bends in the delivery catheter lumen, including an anticipated direction and/or degree of a sharpest bend.

(69) As described herein, a spacer can be adjacent to a puncture component or a predetermined distance proximal of the puncture component. For example, the spacers **250, 350, 450, 550, 650, 750, 850** described with reference to FIGS. **4** through **10**, can be adjacent to or a predetermined distance proximal of the respective puncture component **206, 306, 406, 506, 606, 706, 806**. The distal end **254, 354, 454, 554, 654, 754, 854** of the respective spacer **206, 306, 406, 506, 606, 706, 806** can be adjacent to or a predetermined distance proximal of a respective proximal end **208, 308, 408, 508, 608, 708** of the puncture components **206, 306, 406, 506, 606, 706, 806**.

(70) In some embodiments, a width and/or a length of the spacers **250, 350, 450, 550, 650, 750, 850** can be predetermined based at least in part on a distance between the spacers **250, 350, 450, 550, 650, 750, 850** and the respective pointed sharp distal end **210, 310, 410, 510, 610, 710, 810** of the puncture components **206, 306, 406, 506, 606, 706, 806**. The length can be a longitudinal dimension parallel or substantially parallel to a longitudinal axis of the respective puncture needles **200, 300, 400, 500, 600, 700, 800**. The width can be a lateral dimension which is perpendicularly or substantially perpendicularly to the length, such as perpendicular or substantially perpendicular to the longitudinal axes of the puncture needles **200, 300, 400, 500, 600, 700, 800**. In some embodiments, the width can be directly related to the distance. In some embodiments, the length can be inversely related to the distance. In some embodiments, the length does not depend on the distance between the spacers **250, 350, 450, 550, 650, 750, 850** and the respective pointed sharp distal end **210, 310, 410, 510, 610, 710, 810** of the puncture components **206, 306, 406, 506, 606, 706, 806**.

(71) As shown in FIGS. **4** through **10**, the proximal ends **252, 352, 452, 552, 652, 752, 852** and/or the distal ends **254, 354, 454, 554, 654, 754, 854** can be a planar or substantially planar surface. Alternatively, the spacers **250, 350, 450, 550, 650, 750, 850** can comprise a proximal taper or a distal taper, such as a taper from the lateral portion **256, 356, 456, 556, 656, 756, 856** toward the proximal ends **252, 352, 452, 552, 652, 752, 852** and/or the distal ends **254, 354, 454, 554, 654, 754, 854**, respectively. A taper may include a curved and/or a linear taper. The spacers **250, 350, 450, 550, 650, 750, 850** can comprise a proximal taper and/or a distal taper to facilitate retraction and/or insertion of the spacers **250, 350, 450, 550, 650, 750, 850** through the delivery catheter lumen.

(72) In some embodiments, the spacers **250, 350, 450, 550, 650, 750, 850** can serve as a dilator configured to be inserted into the opening formed at a target tissue site to enlarge the opening. In some embodiments, the width of the spacers **250, 350, 450, 550, 650, 750, 850** can be selected based at least in part on a desired dilation of the opening. In some embodiments, the spacers **250, 350, 450, 550, 650, 750, 850** can comprise a distal taper to facilitate insertion of the spacers **250, 350, 450, 550, 650, 750, 850** into the opening. In some embodiments, the distal ends **254, 354, 454, 554, 654, 754, 854** can be planar or substantially planar to prevent or reduce insertion of the spacers **250, 350, 450, 550, 650, 750, 850** into the opening.

(73) FIG. **11** is a flow diagram of an example of a process **1100** for deploying a puncture needle comprising a spacer as described herein. In block **1102**, the process **1100** involves inserting the puncture needle and an associated spacer into a delivery catheter lumen. The puncture needle can comprise an elongate portion and the spacer can be positioned on a distal portion of the elongate portion. The puncture component can comprise a pointed sharp distal end which can become

lodged within a wall of the delivery catheter lumen if the sharp distal end comes into contact with the wall of the delivery catheter lumen.

(74) In block **1104**, the process **1100** involves advancing the puncture needle through a bend in the delivery catheter lumen. For example, the puncture component can be inserted through a bend in the delivery catheter lumen without the puncture component becoming jammed within the delivery catheter lumen. In block **1106**, the process **1100** involves contacting the spacer with a wall of the delivery catheter lumen so as to prevent contact between the wall and the puncture component of the puncture needle. The spacer can contact the wall to provide space between the delivery catheter lumen wall and the pointed sharp distal end of the puncture component such that the pointed sharp distal end does not contact the wall.

(75) In block **1108**, the process **1100** involves extending the puncture component through an outlet opening of the delivery catheter to puncture tissue at a target site. The target tissue site can be at a target site on any number of internal vessels, channels, organs and/or chambers. In some embodiments, the target site can be on a left atrial wall. In some embodiments, the puncture needle can be positioned within a coronary sinus to enable accessing the left atrial wall from within the coronary sinus. For example, a transfemoral approach can be used to position the delivery catheter into the coronary sinus from the right atrium via the coronary sinus ostium. The puncture needle can then be deployed from the delivery catheter lumen to form an opening on the left atrial wall from within the coronary sinus. In some embodiments, a trans-subclavian or a transjugular approach can be used.

(76) In some embodiments, after the opening is formed, at least a portion of the spacer can be optionally inserted into the opening to enlarge the opening. As described herein, in some embodiments, a width the spacer can be selected based on a desired enlargement of the opening formed in the tissue. In some embodiments, the spacer can be configured to reduce or eliminate insertion into the opening in the target tissue.

(77) As described herein, although the spacer is described as being associated with an elongate portion of a puncture needle, in some embodiments, the spacer can instead be associated with a distal exterior portion of a puncture needle sheath configured to receive the puncture needle. The spacer can be on an exterior portion of the puncture needle sheath, such as a distal exterior portion. After the puncture needle sheath and the puncture needle are positioned at a desired position relative to the target tissue site, the puncture needle sheath can be retracted and/or the puncture needle can be advanced to deploy the puncture needle for piercing the target tissue.

(78) FIG. **12** is a longitudinal cross-sectional view of an example of a puncture needle system **1200**. The puncture needle system **1200** can include a puncture needle **1210** and a protective cover **1250** over the puncture needle **1210**. FIG. **12** shows the puncture needle system **1200** positioned within a lumen **1204** of a delivery catheter **1202**. The puncture needle **1210** can comprise an elongate portion **1212** and a puncture component **1216** at a distal end **1214** of the elongate portion **1212**. The puncture component **1216** can have a pointed sharp distal end **1218** and the protective cover **1250** can extend over the pointed sharp distal end **1218** such that the pointed sharp distal end **1218** can be prevented from contacting a wall **1206** of the delivery catheter lumen **1204**. For example, the protective cover **1250**, such as a distal portion **1252** of the protective cover **1250**, can be maintained over the pointed sharp distal end **1218** while the puncture needle **1210** is advanced through any bends in the delivery catheter lumen **1204** such that the pointed sharp distal end **1218** does not contact the wall **1206**.

(79) The protective cover **1250** can extend over at least a portion of the puncture needle **1210**, including over the puncture component **1216**. The delivery catheter **1202** can be positioned at a desired location. Subsequently, the puncture needle **1210** and the protective cover **1250** can be inserted through the delivery catheter lumen **1204** to position the puncture needle **1210** at or proximate to a target tissue site for forming an opening in the tissue. For example, the puncture component **1216** enclosed within the protective cover **1250** can be advanced through the delivery

catheter lumen **1204** until the puncture component **1216** is positioned at or proximate to the target tissue site. After the puncture component **1216** is positioned at or proximate to the target tissue site, the puncture needle **1210** can be translated distally relative to the protective cover **1250** to allow the puncture component **1216** to pierce the distal portion **1252** of the protective cover **1250**. The unsheathed puncture component **1216** can be used to pierce target tissue for forming the opening in the target tissue. The puncture component **1216** can be extended further, for example extended through a puncture needle outlet opening of the delivery catheter **1202**, such that the puncture component **1216** can be used to puncture tissue at the target site.

(80) FIGS. **13** through **25** show various examples of puncture needles which have a puncture component with a blade edge on a distal edge. The blade edge can be sharp so as to be used to cut through tissue so as to form an opening at a target tissue site. In some embodiments, the blade edge can comprise a convex curvature. The blade edge can be a convexly curved blade edge. The blade edge can include a linear portion between opposing curved corners. As described herein, conventional puncture components typically include a pointed sharp distal end. The pointed sharp distal end of the puncture component can become lodged within a delivery catheter lumen wall when the pointed sharp distal end contacts the wall, such as when the puncture component is advanced through a bend in the delivery catheter lumen. The pointed sharp distal end can prick the delivery catheter lumen wall and stick to the wall, resulting in the puncture needle becoming jammed within the delivery catheter lumen. In contrast, the curvature in the blade edge can prevent the blade edge from becoming stuck against the delivery catheter lumen wall when the blade edge contacts the wall. For example, the convexly curved blade edge and/or the blade edge comprising the linear portion between opposing rounded corners can slide along the wall without sticking to the wall when the blade edge contacts the wall at a bend in the delivery catheter lumen. Puncture needles comprising a blade edge as described herein can thereby be advanceable through one or more bends of a delivery catheter lumen.

(81) FIG. **13** shows an example of a puncture needle **1300** positioned within a delivery catheter **1350**, where the puncture needle **1300** includes a puncture component **1310** with a blade edge **1320**. In some embodiments, the puncture component **1310** can comprise a rounded paddle shape, where the blade edge **1320** is on at least a portion of the rounded distal edge of the paddle. The blade edge **1320** can be a sharp edge configured to pierce tissue. The blade edge **1320** can glide along the wall **1354** of the delivery catheter lumen **1352** if the blade edge **1320** contacts the wall **1354**, without any portion of the blade edge **1320** becoming lodged within the wall **1354**. For example, as the puncture component **1310** is advanced through a bend in the delivery catheter lumen **1354**, the blade edge **1320** can come into contact with the delivery catheter lumen wall **1354** without any portion of the blade edge **1320** sticking to the wall **1354**. The blade edge **1320** can advantageously facilitate insertion of the puncture needle **1300** through bends in the delivery catheter lumen **1352**, without the puncture needle **1300** becoming stuck within the lumen **1352**.

(82) Referring to FIG. **13**, the puncture needle **1300** can comprise an elongate portion **1302** and the puncture component **1310** can be at a distal end **1304** of the elongate portion **1302**. The puncture component **1310** can have a proximal portion **1312** and a distal portion **1314**. The proximal portion **1312** can comprise a proximal end **1316** adjacent to the distal end **1304** of the elongate portion **1302**. The distal portion **1314** can comprise a distal edge **1318** which includes the blade edge **1320**. The distal portion **1314** can comprise a curved edge, such as a convex curve. The distal edge **1318** can comprise the curved edge such that the blade edge **1320** is a curved blade edge. The radius of curvature of the curved edge may or may not be uniform. In some embodiments, the distal portion **1314** can comprise a segment of a circle and/or a segment of an oval. The blade edge **1320** can comprise a segment of a circle and/or a segment of an oval.

(83) FIG. **14A** is a perspective view of the puncture needle **1300**. The puncture component **1310** can have two opposing surfaces **1322**, **1324** which converge along respective distal edges to form the distal edge **1318**. For example, the opposing surfaces **1322**, **1324** can be a top surface and a

bottom surface, respectively, of the puncture component **1310**. The opposing surfaces **1322**, **1324** can be planar or substantially planar. In some embodiments, only a portion of the distal edge **1318** comprises the blade edge **1320**. For example, a portion of the distal edge **1318** can be blunt. In some embodiments, the blade edge **1320** can extend along the entirety or substantially the entirety of the distal edge **1318**. For example, opposing surfaces **1322**, **1324** can converge to form the sharp blade edge **1320**.

(84) Lateral surfaces **1326**, **1328** of the puncture component **1310** can extend between the two opposing surfaces **1322**, **1324**. In some embodiments, the lateral surfaces **1326**, **1328** do not or substantially do not bow and/or curve along a dimension extending between the two opposing surfaces **1322**, **1324**. A width of the puncture component **1310**, such as W.sub.1300, can be a dimension which extends between respective portions of the lateral surfaces **1326**, **1328**. The width W.sub.1300 can be a dimension extending between opposing portions of the lateral surfaces **1326**, **1328**. For example, a width W.sub.1300 can extend between a portion of the lateral surface **1326** and a portion of the lateral surface **1328** opposite the portion of the lateral surface **1326**. A height of the puncture component **1310**, such as H.sub.1300, can be a dimension extending between respective portions of the opposing surfaces **1322**, **1324**. The height can be a dimension extending between opposing portions of the surfaces **1322**, **1324**, between a portion of the top surface **1322** and a portion of the bottom surface **1324** opposite the portion of the top surface **1322**. For example, the height can be a dimension perpendicular or substantially perpendicular to a longitudinal axis L.sub.1300 of the puncture needle **1300**. A width can be a dimension perpendicular or substantially perpendicular to both a height and the longitudinal axis L.sub.1300 of the puncture needle **1300**.

(85) FIG. 14B shows a top-down plan view of the puncture needle **1300**. FIG. 14C shows a side view of the puncture needle **1300**. Referring to FIG. 14B, the proximal portion **1312** can have a width which increases along a direction extending from the proximal end **1316** toward the distal portion **1314**. In some embodiments, the top surface **1322** and the bottom surface **1324** can be planar. In some embodiments, the top surface **1322** and the bottom surface **1324** are parallel or substantially parallel to one another. In some embodiments, a cross section of the proximal portion **1312** taken along a plane parallel or substantially parallel to the top surface **1322** or the bottom surface **1324** can comprise a trapezoidal shape. For example, a cross section of the proximal portion **1312** taken along a plane parallel or substantially parallel to the top surface **1322** or the bottom surface **1324** and which includes a portion of the lateral surface **1326** and a portion of the lateral surface **1328** opposite that of the lateral surface **1326** can comprise a trapezoidal shape. The trapezoidal shape can have a pair of opposing parallel sides where a shorter of the parallel sides is positioned proximally and a longer of the parallel sides is positioned distally. A cross section of the distal portion **1314** taken along the same plane can comprise a segment of a circle. For example, the cross section of the distal portion **1314** can comprise a semi-circle, the blade edge **1320** being on at least a portion of the semi-circle. The blade edge **1320** can be on a distal portion of the segment of the circle, and opposing portions of the segment proximal of the blade edge **1320** can be blunt portions. A portion of the puncture component **1310** can comprise an increasing width along a direction extending from the distal edge **1318** toward a proximal portion **1312** so as to facilitate use of the puncture component **1310** as a dilator to enlarge an opening formed using the blade edge **1320**.

(86) In some embodiments, the rounded edge of the distal portion **1314**, such as the semi-circle shape, can facilitate use of the puncture component **1310** to enlarge the opening formed using the blade edge **1320**. The rounded edge can facilitate gradual dilation of the opening. As the enlargement can be performed using one or more blunt portions of the puncture component **1310**, atraumatic incremental enlargement can be achieved. For example, enlarging the opening can comprise inserting the blunt portions on the segment of the semi-circle of the distal portion **1314** into the opening to atraumatically enlarge the opening.

(87) In some embodiments, the puncture component **1310** comprising a wider width at a portion

proximal of the blade edge **1320** can reduce or prevent contact between the blade edge **1320** and a wall of a delivery catheter lumen. For example, the lateral surfaces **1326**, **1328** can contact the wall such that desired space can be maintained between the wall and the sharp blade edge **1320**.

(88) Referring to FIG. **14C**, the proximal portion **1312** can comprise a portion having a uniform or substantially uniform height. In some embodiments, a cross section of the proximal portion **1312** taken along a plane perpendicular or substantially perpendicular to the top surface **1322** or the bottom surface **1324** can comprise a rectangular shape. For example, a cross section of the proximal portion **1312** taken along a plane which includes opposing portions of the top surface **1326** and bottom surface **1328**, a portion of the top surface **1326** and a portion of the bottom surface **1328** opposite that of the top surface **1326**, and which extends along a direction extending from the proximal end **1316** to the distal portion **1314**, can comprise a rectangular shape. The distal portion **1314** can have a height which decreases along a direction extending from the proximal portion **1312** to the distal edge **1318**. For example, a cross section of the distal portion **1314** taken along a plane which includes opposing portions of the top surface **1322** and bottom surface **1324**, and which extends along a direction extending from the proximal portion **1312** to the distal edge **1318** can comprise a triangular shape.

(89) FIG. **15** is a perspective view of another example of a puncture needle **1400** which includes a puncture component **1410** with a blade edge **1420**. The puncture needle **1400** can have lateral surfaces **1426**, **1428** which bow and/or curve outward. Other features of the puncture needle **1400** can be the same as or similar to those of the puncture needle **1300** described with reference to FIGS. **13** and **14**. For example, the puncture component **1410** can comprise a rounded paddle shape, where the blade edge **1420** is on at least a portion of the rounded distal edge of the paddle, and where lateral surfaces **1426**, **1428** of the puncture component **1410** comprise one or more curvatures. The puncture component **1410** can have a top planar or substantially planar surface **1422** and a bottom planar or substantially planar surface **1424** which converge along respective distal edges to form the distal edge **1418**. The blade edge **1420** can be on the distal edge **1418**. The blade edge **1420** can extend along the entirety or substantially the entirety of the distal edge **1418**, or can extend only along a portion of the distal edge **1418**. The lateral surfaces **1426**, **1428** can extend between the top surface **1422** and the bottom surface **1424**.

(90) A width of the puncture component **1410** can be a dimension which extends between opposing portions of the lateral surfaces **1426**, **1428**, a portion of the lateral surface **1426** and a portion of the lateral surface **1428** opposite that of the lateral surface **1426**. A height of the puncture component **1410** can be a dimension extending between opposing portions of the top surface **1422** and bottom surface **1424**, for example, being perpendicular or substantially perpendicular to a longitudinal axis of the puncture needle **1400**. The width can be perpendicular or substantially perpendicular to both the height and the longitudinal axis of the puncture needle **1400**.

(91) In some embodiments, the top surface **1422** and the bottom surface **1424** can be planar. In some embodiments, the top surface **1422** and the bottom surface **1424** are parallel or substantially parallel to one another. A cross section of the proximal portion **1412** taken along a plane parallel or substantially parallel to the top surface **1422** or the bottom surface **1424** can comprise a trapezoidal shape. For example, a cross section of the proximal portion **1412** taken along a plane parallel or substantially parallel to the top surface **1422** or the bottom surface **1424** and which includes opposing portions of the lateral surfaces **1426**, **1428** can comprise a trapezoidal shape. A width of the proximal portion **1412** can increase along a direction extending from the proximal end **1416** toward the distal portion **1414**. A cross section of the distal portion **1414** taken along the same plane can comprise a semi-circle, the blade edge **1420** being on at least a portion of the semi-circle. The blade edge **1420** can be a convexly curved blade edge. The proximal portion **1412** can comprise a portion having a uniform or substantially uniform height. For example, a cross section of the proximal portion **1412** taken along a plane which includes opposing portions of the top surface **1426** and bottom surface **1428**, and which extends along a direction extending from the

proximal end **1416** to the distal portion **1414**, can comprise a rectangular shape. The distal portion **1414** can have a height which decreases along a direction extending from the proximal portion **1412** to the distal edge **1418**. For example, a cross section of the distal portion **1414** taken along the plane which includes opposing portions of the top surface **1422** and bottom surface **1424**, and which extends along a direction extending from the proximal end **1416** to the distal portion **1414**, can comprise a triangular shape.

(92) As described herein, the lateral surfaces **1426**, **1428** can bow and/or curve outward. Each of the lateral surfaces **1426**, **1428** can comprise one or more convex curvatures along a direction extending between the top surface **1422** and the bottom surface **1424**, for example perpendicular to or substantially perpendicular to the top surface **1422** and the bottom surface **1424**. For example, the lateral surfaces **1426**, **1428** can be rounded. The rounded lateral surfaces **1426**, **1428** can facilitate insertion and/or retraction of the puncture needle **1400** through a delivery catheter lumen. The one or more curvatures can reduce friction between the puncture component **1410** and the delivery catheter lumen wall. In some embodiments, the rounded lateral surfaces **1426**, **1428** can facilitate insertion of the puncture component **1410** into a tissue opening formed using the blade edge **1420** to provide atraumatic enlargement of the opening.

(93) FIGS. **16** through **23** are top-down plan views of additional examples of puncture components which comprise a blade edge. FIGS. **16** to **19** are top-down plan views of some examples of puncture components, or portions thereof, which comprise a curved blade edge. The puncture components **1500**, **1600**, **1700**, **1800** shown in FIGS. **16** to **19** can each have a respective blade edge **1508**, **1608**, **1708**, **1808** which comprises a curve. Each of the puncture components **1500**, **1600**, **1700**, **1800** can have a respective proximal end **1502**, **1602**, **1702**, **1802**, and a respective distal edge **1504**, **1604**, **1704**, **1804**. In some embodiments, the puncture components **1500**, **1600**, **1700**, **1800**, shown can be only a portion of a puncture component, for example the respective proximal ends **1502**, **1602**, **1702**, **1802** being coupled to an intermediate member which is in turn coupled to an elongate portion of a puncture needle. Each of the distal edges **1504**, **1604**, **1704**, **1804** can comprise a curved portion, such as a convex curvature. The curvature can have a uniform or non-uniform radius of curvature. For example, each of the distal edges **1504**, **1604**, **1704**, **1804** can be a convexly curved edge, including a convexly curved edge which comprises a segment of a circle and/or an oval. The blade edges **1508**, **1608**, **1708**, **1808** can be on the respective distal edge **1504**, **1604**, **1704**, **1804**. For example, each of the blade edges **1508**, **1608**, **1708**, **1808** can be a convexly curved blade edge, including a blade edge which is a segment of a circle and/or an oval. In some embodiments, the blade edges **1508**, **1608**, **1708**, **1808** can extend along an entirety of a respective distal edge **1504**, **1604**, **1704**, **1804**. In some embodiments, the blade edges **1508**, **1608**, **1708**, **1808** can extend along only a portion of the corresponding distal edge **1504**, **1604**, **1704**, **1804**. For example, a portion of the distal edges **1504**, **1604**, **1704**, **1804** can be blunt.

(94) A width of a puncture component can increase, decrease or remain the same or substantially the same along a direction extending from a proximal end to a distal edge. Each of the puncture components **1500**, **1600**, **1700**, **1800** can have a respective pair of lateral sides **1506**, **1606**, **1706**, **1806** extending between the corresponding proximal end **1502**, **1602**, **1702**, **1802** and distal edge **1504**, **1604**, **1704**, **1804**. The width can be a dimension extending between opposing portions of the lateral sides **1506**, **1606**, **1706**, **1806**. A width of the puncture component **1500** can decrease along a direction extending from the distal edge **1504** to the proximal end **1502**. A width of the puncture component **1600** can remain the same or substantially the same along at least a portion of a dimension extending from the proximal end **1602** to the distal edge **1604**. A width of the puncture components **1700**, **1800** can increase along a direction extending from the respective distal edge **1704**, **1804** to the respective proximal end **1702**, **1802**. In some embodiments, the increase width along a direction extending from the respective distal edge **1704**, **1804** to the respective proximal end **1702**, **1802** can facilitate use of the puncture components **1700**, **1800** as a dilator to enlarge an opening formed at a target tissue site.

(95) Referring to FIGS. **16**, **17** and **18**, the lateral sides **1506**, **1606**, **1706** can comprise a linear or substantially linear portion. Referring to FIG. **19**, the pair of lateral sides **1806** extending between the proximal end **1802** and the distal edge **1804** of the puncture component **1800** can comprise a curvature. For example, the pair of lateral sides **1806** can comprise a convex curvature. As shown in FIG. **16**, in some embodiments, the lateral sides **1506** of the puncture component **1500** can meet the distal edge **1504** at rounded corners **1510**. The rounded corners **1510** can facilitate insertion of the puncture component **1500** through a delivery catheter lumen. Referring to FIG. **17**, the distal edge **1604** can comprise a segment of a circle, such as a semi-circle. For example, the blade edge **1608** can comprise a semi-circle. Referring to FIGS. **18** and **19**, the distal edges **1704**, **1804** of the puncture components **1700**, **1800** can comprise a segment of an oval. For example, the blade edges **1708**, **1808** can each comprise a segment of an oval.

(96) FIGS. **20** to **23** are top-down plan views of some examples of puncture components with a blade edge which includes a linear or substantially linear portion between opposing curvatures, such as between opposing rounded corners. The puncture components **1900**, **2000**, **2100**, **2200** shown in FIGS. **20** to **23** can each comprise a respective linear or substantially linear blade edge **1908**, **2008**, **2108**, **2208**. Each of the puncture components **1900**, **2000**, **2100**, **2200** can have a respective proximal end **1902**, **2002**, **2102**, **2202**, and a respective distal edge **1904**, **2004**, **2104**, **2204**. Each of the distal edges **1904**, **2004**, **2104**, **2204** can comprise a linear or substantially linear portion. For example, each of the distal edges **1904**, **2004**, **2104**, **2204** can be linear or substantially linear edges. The blade edges **1908**, **2008**, **2108**, **2208** can each be on the respective distal edge **1904**, **2004**, **2104**, **2204**. For example, each of the blade edges **1908**, **2008**, **2108**, **2208** can be a linear or substantially linear blade edge. In some embodiments, the blade edges **1908**, **2008**, **2108**, **2208** can extend along an entirety of respective distal edges **1904**, **2004**, **2104**, **2204**, the respective distal edges **1904**, **2004**, **2104**, **2204** being the blade edges **1908**, **2008**, **2108**, **2208**.

(97) Each of the puncture components **1900**, **2000**, **2100**, **2200** can have a pair of respective lateral sides **1906**, **2006**, **2106**, **2206** extending between the respective proximal end **1902**, **2002**, **2102**, **2202**, and the respective distal edge **1904**, **2004**, **2104**, **2204**. Referring to FIGS. **20**, **22** and **23**, the lateral sides **1906**, **2106**, **2206** of the puncture components **1900**, **2000**, **2100** can each comprise a linear or substantially linear portion. Referring to FIG. **21**, the pair of lateral sides **2006** of the puncture component **2000** extending between the proximal end **2002** and the distal edge **2004** can comprise a curvature. For example, the pair of lateral sides **2006** can comprise a convex curvature.

(98) Each of the pairs of lateral sides **1906**, **2006**, **2106**, **2206** can meet the respective distal edges **1904**, **2004**, **2104**, **2204** at rounded corners **1910**, **2010**, **2110**, **2210**. In some embodiments, the rounded corners **1910**, **2010**, **2110**, **2210** can be a part of the blade edges **1908**, **2008**, **2108**, **2208**. The rounded corners **1910**, **2010**, **2110**, **2210** can be sharp to facilitate tissue puncture at the target site. In some embodiments, the rounded corners **1910**, **2010**, **2110**, **2210** can facilitate insertion of the puncture component **1900**, **2000**, **2100**, **2200** through a delivery catheter lumen, such as to enable gliding of the puncture components **1900**, **2000**, **2100**, **2200** along a wall of the delivery catheter lumen if contact occurs between the puncture components **1900**, **2000**, **2100**, **2200**.

(99) A width of a puncture component comprising a linear or substantially linear blade edge can increase, decrease or remain the same or substantially the same along a direction extending from a distal edge to a proximal end. The width can be a dimension extending between opposing portions of the lateral sides **1906**, **2006**, **2106**, **2206**. A width of the puncture components **1900**, **2000** of FIGS. **20** and **21** can decrease along at least a portion of a dimension extending from the distal edge **1604** to the proximal end **1602**. A width of the puncture component **2200** of FIG. **23** can be the same or substantially the same along a direction extending from the distal edge **2204** to the proximal end **2202**. A width of the puncture component **2100** of FIG. **22** can increase along a direction extending from the distal edge **2104** to the proximal end **2102**. In some embodiments, the increase in width along a direction extending from the distal edge **2104** to the proximal end **2102** can facilitate use of the puncture component **2100** as a dilator to enlarge an opening formed at a

target tissue site.

(100) FIGS. **24** and **25** are longitudinal cross-sectional views of examples of puncture components **2300**, **2400**, respectively. The puncture components **2300**, **2400** can comprise respective proximal ends **2302**, **2402** and respective distal ends **2304**, **2404**. Two opposing surfaces can extend between the ends. For example, a first top surface **2306** and a first bottom surface **2308** can extend between the proximal end **2302** and the distal end **2304**. A second top surface **2406** and a second bottom surface **2408** can extend between the proximal end **2402** and the distal end **2404**.

(101) The top surfaces **2306**, **2406** and the bottom surfaces **2308**, **2408** can meet along respective distal edges which include the distal ends **2304**, **2404**. The blade edges of the puncture components **2300**, **2400** can be a part of the distal edges. For example, the blade edges can extend into and/or out of the page on which FIGS. **23** and **24** are shown.

(102) The top surfaces **2306**, **2406** and the bottom surfaces **2308**, **2408** can have one or more opposing portions which slope toward one another. Referring to FIG. **23**, the first top surface **2306** can comprise a proximal portion **2310** which is parallel or substantially parallel to an opposing portion of the first bottom surface **2308**, such as a proximal portion **2314** of the first bottom surface **2308**. The first top surface **2306** can comprise a first top distal portion **2312** and the first bottom surface **2308** can comprise a first bottom distal portion **2316**. The first top distal portion **2312** and the first bottom distal portion **2316** can slope toward one another in a direction extending from the first proximal end **2302** to the first distal end **2304**. The first top distal portion **2312** and the first bottom distal portion **2316** can slope toward one another and meet at the first distal end **2304**. A slope of the first top distal portion **2312** and a slope of the first bottom distal portion **2316** can be the same or different. For example, the first top distal portion **2312** and the first bottom distal portion **2316** can slope toward one another to form the distal edge which includes the blade edge of the puncture component **2300**. A slope of the first top distal portion **2312** and a slope of the first bottom distal portion **2316** can be the same or different.

(103) Referring to FIG. **25**, the second top surface **2406** can comprise a second top proximal portion **2410** which is parallel or substantially parallel to an opposing portion of the second bottom surface **2408**, such as the second bottom proximal portion **2416**. The second top surface **2406** can comprise a second top central portion **2412** and a second top distal portion **2414**. The second bottom surface **2408** can comprise a second bottom central portion **2418** and a second bottom distal portion **2420**. The second top central portion **2412** and the second bottom central portion **2418** can slope toward one another in a direction extending from the second proximal end **2402** to the second distal end **2404**. A slope of the second top central portion **2412** and a slope of the second bottom central portion **2418** can be the same or different. The second top distal portion **2414** and the second bottom distal portion **2418** can slope toward one another in a direction extending from the second proximal end **2402** to the second distal end **2404**. The second top distal portion **2414** and the second bottom distal portion **2418** can slope toward one another and meet at the second distal end **2404**. For example, the second top distal portion **2414** and the second bottom distal portion **2418** can slope toward one another to form the distal edge which includes the blade edge of the puncture component **2400**. A slope of the second top distal portion **2416** and a slope of the second bottom distal portion **2418** can be the same or different. A slope of the second top central portion **2412** can be different from that of the second top distal portion **2414**. A slope of the second bottom central portion **2418** can be different from that of the second bottom distal portion **2420**.

(104) As described herein, in some embodiments, one or more portions of a puncture component can be inserted into a tissue opening to enlarge the opening formed in the tissue. For example, a height, a width and/or a degree of taper of one or both of the puncture components **2300**, **2400** can be selected based on a desired enlargement of the opening. A height can be a dimension extending between a portion of the top surfaces **2306**, **2406** and a portion of the respective bottom surfaces **2308**, **2408** opposite that of the top surfaces **2306**, **2406**. A width can be a dimension perpendicular or substantially perpendicular to the height, for example extending into and/or out of the page.

(105) In some embodiments, the top surfaces **2306**, **2406** and the bottom surfaces **2308**, **2408** can be planar or substantially planar. In some embodiments, the top surfaces **2306**, **2406** and the bottom surfaces **2308**, **2408** can comprise one or more curvatures, including a convex curvature. For example, the puncture components **2300**, **2400** can comprise a generally conical shape.

(106) FIG. **26** is a process flow diagram of an example of a process **2600** for deploying a puncture needle comprising a puncture component with a blade edge as described herein. In block **2602**, the process **2600** involves inserting a puncture needle into a delivery catheter lumen. The puncture needle can include a puncture component which has a blade edge configured for piercing tissue. The blade edge can have any one of the configurations as described herein. In some embodiments, the blade edge can be a convexly curved blade edge. In some embodiments, the blade edge can include a linear or substantially linear blade edge between opposing curved corners.

(107) In block **2604**, the process **2600** involves advancing the puncture needle through a bend in the delivery catheter lumen. The puncture component can be advanced through the bend without sticking to a wall of the delivery catheter lumen. For example, the blade edge can facilitate sliding of the puncture component along the wall of the delivery catheter lumen. In some embodiments, a portion of the puncture component proximal of the blade edge can have a width wider than a width of the blade edge to facilitate maintaining a desired space between the blade edge and the wall of the delivery catheter lumen, so as to reduce or avoid contact between the blade edge and the wall.

(108) In block **2606**, the process **2600** involves extending the puncture component through an outlet opening of the delivery catheter. After the puncture needle is advanced to a desired location, the puncture component can be deployed. In block **2608**, the process **2600** involves using the blade edge to puncture tissue at a target tissue site to form an opening in the tissue. As described herein, the target tissue site can be at a target site on any number of internal vessels, channels, organs and/or chambers. In some embodiments, the target site can be on a left atrial wall. In some embodiments, the puncture needle can be positioned within a coronary sinus to enable accessing the left atrial wall from within the coronary sinus. For example, a transfemoral approach can be used to position the delivery catheter into the coronary sinus from the right atrium via the coronary sinus ostium. The puncture needle can then be deployed from the delivery catheter lumen to form an opening on the left atrial wall from within the coronary sinus. In some embodiments, a trans-subclavian or a transjugular approach can be used.

(109) In some embodiments, after the opening is formed, the puncture component comprising the blade edge can be optionally inserted further into the opening to enlarge the opening. As described herein, a dimension and/or a shape of the puncture component can be selected based on a desired enlargement of the opening formed in the tissue. For example, a width of the puncture component can increase along a direction extending from the blade edge toward a proximal portion of the puncture component to facilitate enlargement of the opening. As described herein, gradual dilation can be achieved using, for example, puncture components comprising a rounded shape. As the enlargement can be performed using one or more blunt portions of the puncture component, atraumatic incremental enlargement can also be achieved.

ADDITIONAL EMBODIMENTS

(110) Depending on the embodiment, certain acts, events, or functions of any of the processes or algorithms described herein can be performed in a different sequence, may be added, merged, or left out altogether. Thus, in certain embodiments, not all described acts or events are necessary for the practice of the processes.

(111) Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is intended in its ordinary sense and is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more

embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous, are used in their ordinary sense, and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y and Z,” unless specifically stated otherwise, is understood with the context as used in general to convey that an item, term, element, etc. may be either X, Y or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y and at least one of Z to each be present.

(112) It should be appreciated that in the above description of embodiments, various features are sometimes grouped together in a single embodiment, Figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of one or more of the various inventive aspects. This method of disclosure, however, is not to be interpreted as reflecting an intention that any claim require more features than are expressly recited in that claim. Moreover, any components, features, or steps illustrated and/or described in a particular embodiment herein can be applied to or used with any other embodiment(s). Further, no component, feature, step, or group of components, features, or steps are necessary or indispensable for each embodiment. Thus, it is intended that the scope of the inventions herein disclosed and claimed below should not be limited by the particular embodiments described above, but should be determined only by a fair reading of the claims that follow.

(113) It should be understood that certain ordinal terms (e.g., “first” or “second”) may be provided for ease of reference and do not necessarily imply physical characteristics or ordering. Therefore, as used herein, an ordinal term (e.g., “first,” “second,” “third,” etc.) used to modify an element, such as a structure, a component, an operation, etc., does not necessarily indicate priority or order of the element with respect to any other element, but rather may generally distinguish the element from another element having a similar or identical name (but for use of the ordinal term). In addition, as used herein, indefinite articles (“a” and “an”) may indicate “one or more” rather than “one.” Further, an operation performed “based on” a condition or event may also be performed based on one or more other conditions or events not explicitly recited.

(114) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(115) The spatially relative terms “outer,” “inner,” “upper,” “lower,” “below,” “above,” “vertical,” “horizontal,” and similar terms, may be used herein for ease of description to describe the relations between one element or component and another element or component as illustrated in the drawings. It be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the drawings. For example, in the case where a device shown in the drawing is turned over, the device positioned “below” or “beneath” another device may be placed “above” another device. Accordingly, the illustrative term “below” may include both the lower and upper positions. The device may also be oriented in the other direction, and thus the spatially relative terms may be interpreted differently depending on the orientations.

(116) Unless otherwise expressly stated, comparative and/or quantitative terms, such as “less,” “more,” “greater,” and the like, are intended to encompass the concepts of equality. For example, “less” can mean not only “less” in the strictest mathematical sense, but also, “less than or equal to”.

Claims

1. A puncture needle comprising: an elongate portion; and a puncture component associated with a distal end of the elongate portion, the puncture component comprising a proximal portion, an intermediate portion, and a distal portion, the distal portion comprising a distal edge, the distal edge forming a semi-circle, the semi-circle comprising a central blade edge, opposing portions of the semi-circle proximal to the blade edge being blunt, the proximal portion and the distal portion having a height that tapers and the intermediate portion having a constant height, and the blade edge being configured to puncture tissue at a target site; and wherein the puncture component is configured to be advanced through a lumen of a delivery catheter without the blade edge sticking to a wall of the lumen at a bend in the delivery catheter.
 2. The puncture needle of claim 1, wherein the proximal portion of the puncture component has a width narrower than a width of the blade edge.
 3. The puncture needle of claim 1, wherein the proximal portion of the puncture component has a width the same as a width of the blade edge.
 4. The puncture needle of claim 1, wherein the proximal portion of the puncture component has a height larger than that of the distal portion of the puncture component.
 5. A puncture needle comprising: an elongate portion; and a puncture component coupled to a distal end of the elongate portion, the puncture component having a top planar surface and a bottom opposing planar surface and lateral surfaces extending between respective edges of the top and bottom opposing planar surfaces, and the puncture component comprising: a proximal portion, a cross section of the proximal portion taken along a plane including opposing portions of the lateral surfaces comprising a trapezoidal shape, a proximal parallel side of the trapezoidal shape being shorter than a distal parallel side of the trapezoidal shape; and a distal portion, wherein corresponding portions of the top planar surface and bottom opposing planar surface of the distal portion are converging along respective distal edges to form a convexly curved distal edge, the convexly curved distal edge comprising a blade edge, the blade edge is on a distal portion of the convexly curved distal edge, and opposing portions of the convexly curved distal edge proximal of the blade edge being blunt portions.
 6. The puncture needle of claim 5, wherein a width of the proximal portion increases along a direction extending from a proximal end of the puncture component toward the distal portion.
 7. The puncture needle of claim 5, wherein the lateral surfaces do not comprise a bow along a dimension extending between the top and bottom opposing planar surfaces.
 8. The puncture needle of claim 5, wherein the blade edge comprises a segment of an oval.
 9. The puncture needle of claim 5, wherein the blade edge comprises a segment of a circle.
 10. The puncture needle of claim 9, wherein the blade edge comprises a semi-circle.
 11. The puncture needle of claim 5, wherein the proximal portion comprises a uniform thickness.
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